



Mobile Manipulators (Image Courtesy: Handling Concepts, UK)

agentic AI architectures are convoluted, requiring diverse and multiple models, sophisticated retrieval-augmented generation stacks, advanced data architectures, and niche expertise. It suggests that companies should prefer working with AI service providers and systems integrators rather than attempting to build advanced agentic architectures themselves.

• **Small language models (SLMs).** For some, small could be big! Unlike large language models (LLMs), which are trained with loads of data to achieve multi-tasking, SLMs are trained in specific domains with specific databases.

SLM parameters range from a few million to a few billion, unlike LLMs with hundreds of billions or even trillions of parameters. As a result, they require less resources and investment, have a quicker response time, and offer better security. They can be deployed for customer service, content moderation, translation, and such tasks.

Small agent models are AI agents that use SLMs. They have a task-oriented design with reasonably good knowledge of the specific domain. Basically, a small AI agent

is not a jack of all trades—but a master of one! Small agent models can potentially be trained as healthcare assistants, wealth managers, and e-commerce agents.

Small agent models are not as smart as regular AI agents discussed above—they can interact only using text and can solve only specific problems within the contexts that they have been trained in. However, SLMs and small agent

models can work in resource-constrained environments such as edge devices or areas without connectivity. If a small enterprise requires only limited but focused capabilities with low initial investment, SLMs will help them board the AI train in 2025!

• **Customised LLMs.** Large enterprises favour tailor-made large-language models (LLMs), which combine the capabilities of generative AI with proprietary enterprise data to achieve tasks specific to the organisation. At last, the data that organisations have collected at the edge for several years (without really knowing what for) will be put to actual good use!

• **AI governance platforms.** With power comes great responsibility! We see AI being misused all over—from deepfakes to security breaches, we have seen it all. Companies that are serious about their AI deployments will have to implement an AI governance framework to manage the legal, ethical and operational aspects of their AI systems—including tasks like fostering trust, preventing fraud and disinformation, data governance, continuous risk scoring, contextual

awareness and even post-quantum cryptography.

• **Strategic planning and implementation.** As the initial hype clears, enterprises are likely to realise that AI is a long-term investment which requires proper strategic planning and execution. According to some analysts, companies that invested heavily in it in 2024, expecting instant returns on investment (RoI), are likely to scale back their investments prematurely. On the other hand, those serious about their AI initiatives are likely to chart out a clear plan of action for AI deployment, which also considers trust and security factors.

• **Invisible intelligence.** AI and IoT will likely come together to create an invisible thread of intelligence across spaces. While IoT collects data at every opportunity, AI has the power to instantly convert it into actionable insights, making the duo a formidable team! We will see this trend coming to life in small spaces in 2025, and over the years, it is likely to grow bigger—spanning whole neighbourhoods and towns as well.

• **Pervasive influence.** AI is likely to be the key influencer in almost every tech space ranging from robotics and automation to blockchain and telecommunications.

The world is also watching how India makes its AI moves! John Chambers, chairman of the US-India Strategic Partnership Forum (USISPF), whose predictions are widely respected by the tech industry, remarks in his predictions for 2025 that countries that lead in AI will see the fastest gross domestic product (GDP) growth as well as a significant improvement in standard of living and defence capabilities. He further states that he sees the US and India emerging as the global frontrunners. “I believe the odds of this happening are better than 50/50, considering the amount of innovation